Forward diffusion process

$$q(\zeta^k \mid \zeta^{k-1}) = \mathcal{N}\left(\zeta^k; \sqrt{\alpha_k}\,\zeta^{k-1}, (1-\alpha_k)I\right)$$

Historical Scenarios

ζ^0 … ζ^{k-1} —*add noise*→ ζ^k … ζ^K

Gaussian noise

approximately $\mathcal{N}(0, I)$

Learned reverse denoising process

$$p_\theta(\zeta^{k-1} \mid \zeta^k, c) = \mathcal{N}\left(\zeta^{k-1}; \mu_\theta(\zeta^k, c, k), \Sigma_\theta(\zeta^k, c, k)\right)$$

Generated Scenarios

ζ^0 … ζ^{k-1} ←*denoise*— ζ^k … ζ^K

Gaussian initialization

$\zeta^K \sim \mathcal{N}(0, I)$

One reverse denoising step

ζ^k

diffusion step k

condition c (optional)

Transformer denoiser

reverse mean $\mu_\theta(\zeta^k, c, k)$

Gaussian reverse step

$\mathcal{N}(\mu_\theta, \Sigma_\theta)$

ζ^{k-1}

Optimization-aware training at step k

Historical minibatch $\{\zeta_0\}$

Gaussian corruption at k

Denoiser and reconstruction ε_θ

stochastic program $x(\hat{\zeta}_0, c)$

Empirical decision regret $J_{\text{hist}}(x) - J^*_{\text{hist}}$

Historical optimum x^*_{hist}, J^*_{hist}

Noise and clean-data targets

(from historical minibatch at step k)

Noise-prediction loss $\mathcal{L}^k_{\text{noise}}(\theta)$

Reconstruction loss $\mathcal{L}^k_{\text{recon}}(\theta)$

Optimization loss $\mathcal{L}^k_{\text{opt}}(\theta)$

$$\mathcal{L}^k(\theta) = \lambda_{\text{noise}}\mathcal{L}^k_{\text{noise}}(\theta) + \lambda_{\text{recon}}\mathcal{L}^k_{\text{recon}}(\theta) + \lambda_{\text{opt}}\mathcal{L}^k_{\text{opt}}(\theta)$$

Backpropagate through the decision and reconstruction to update θ

Solid: computation

Dashed: gradient update